Dallas International School

# Test 7

**Topics 4.1 – 4.2 – 4.3**

**11th IB – Math AA**

**For SL & HL Students**

**EXERCISE 1**

Consider the following set of ordered data.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 27 | 27 | 31 | 32 | 33 | 37 | 41 | 46 | 54 | 57 |

**(a)** Write down

**(a.i)** the mode; [1]

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**(a.ii)** the range; [1]

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**(a.iii)** the median. [1]

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**(b)** Find the interquartile range. [3]

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**EXERCISE 2**

The following table shows the decrease in cholesterol levels in 6 volunteers after a two week trial of special diet and exercise.

| *Volunteer* | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| *Decrease in cholesterol* | 0.8 | 0.6 | 0.7 | 0.8 | 0.4 | 2.8 |

**(a)** Find the standard deviation of the data. [2]

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**(b)** Recalculate the standard deviation with the outlier removed. [2]

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**(c)** Discuss the effect of an extreme value on the standard deviation. [2]

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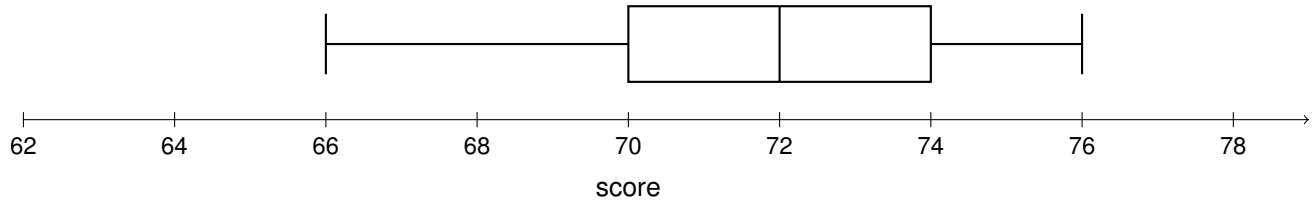
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**EXERCISE 3**

The scores achieved by 80 golfers in a competition are summarized in the following box-and-whisker diagram.




**(a)** Find the interquartile range. [2]

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**(b)** Find the number of golfers that scored between 70 and 74. [2]

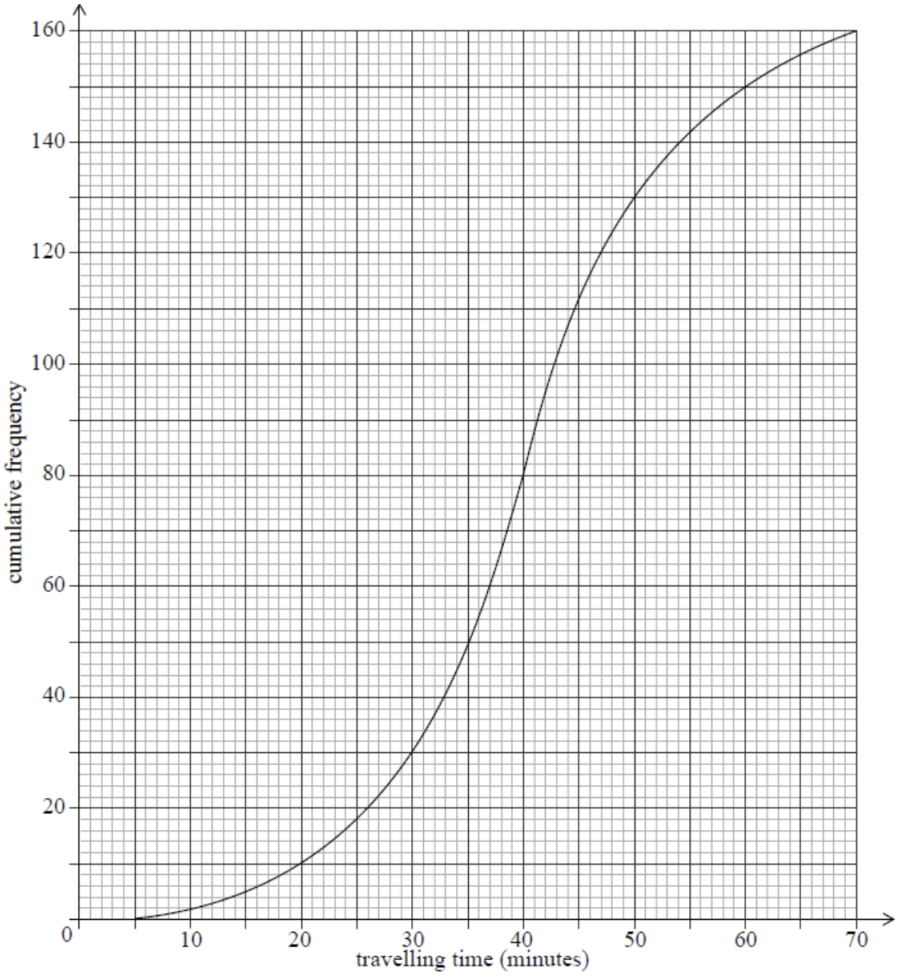
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**(c)** How is this data distributed? Justify your answer. [2]

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**EXERCISE 4**

A company surveys its employees on how long they spend travelling to work each day. The results are shown in the following cumulative frequency diagram.




**(a)** Find the median number of minutes spent travelling to work. [2]

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**(b)** Find the number of employees whose travelling time is within 15 minutes of the median. [3]

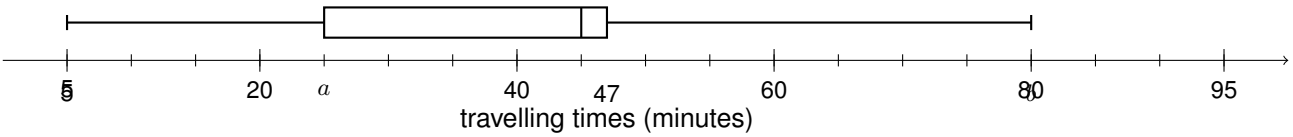
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Only 10% of the employees spent more than $k$ minutes travelling to work.

**(c)** Find the value of $k$. [3]

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The results of the survey can also be displayed on the following box-and-whisker diagram.




**(d)** Write down the value of $b$. [1]

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**(e.i)** Find the value of $a$. [2]

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**(e.ii)** Hence, find the interquartile range. [2]

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**(f)** Travelling times of less than $p$ minutes are considered outliers. Find the value of $p$. [2]

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**EXERCISE 5**

A ballet company performs *The Sleeping Beauty* every year. Last year they gave a total of 60 performances at their theatre which has a maximum capacity of 800. The number of tickets sold, $n$, at each performance is shown in the following frequency table.

| **Number of tickets sold,** $n$ | **Number of performances** |
|---|---|
| $0 < n \leq 200$ | 3 |
| $200 < n \leq 400$ | $p$ |
| $400 < n \leq 600$ | 18 |
| $600 < n \leq 800$ | 30 |

**(a.i)** Find the value of $p$. [1]

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**(a.ii)** Write down the modal class. [1]

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Last year 36 000 tickets were sold to *The Sleeping Beauty*.

**(b.i)** Use the cumulative frequency curve to estimate the median number of tickets sold. [1]

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**(b.ii)** Estimate the number of performances where at least 80% of the tickets were sold. [3]

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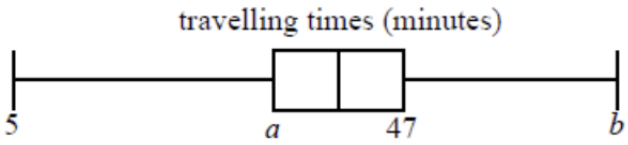
After a performance, the company decides to conduct a survey to obtain feedback from the audience.

**(c.i)** State one disadvantage of the company surveying only the first 5% of the audience as they leave the theatre. [1]

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The following box-and-whisker diagram displays the amount spent by the audience at the souvenir shop when they attended the performance.




**(d.i)** Estimate the number of people who spent between \$3 and \$25. [2]

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**(d.ii)** Half the audience spent less than \$$a$. Estimate the value of $a$. [1]

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